

# PRASHANT RAWAT

Machine Learning Engineer, Celusion Technologies  
M.Tech, IIIT-Allahabad

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## EXPERIENCE

### Celusion Technologies

📅 Sep'21-Present

Role: Team lead and contributor.

- OCR of Indian identity cards (PAN, Aadhaar card) and Philippines Driving License.
- Face matching to check for duplicate users in banking software.
- Face liveness for user verification in banking software.

### Central Research Laboratory-B.E.L (CRL-B.E.L)

📅 Jun'19 - Sep'21

Role: Team lead or Individual contributor (IC).

- IC on Weapon Detection from Real-time Camera feed, Radar feed and electric fence feed for border security.
- Lead and contributor on Coastal Surveillance System to detect unauthorized ships.
- Lead and contributor on Air Traffic Management System.
- IC on implementation of RTI-DDS (Data Distribution Service) for inter-module communication across multiple projects.

### IIIT-Allahabad

📅 Aug'17 - May'19

Role: Teaching Assistant.

- Taught lectures and lab sessions on "Image & Video processing", "Data Mining", "Machine Learning Tools & Technologies" and "Computational Intelligence".

### R.W.X Technology

📅 Jun'15 - Jul'15

Role: Internship

- Semi-Automated FAQ-Retrieval System.
- Hand Gesture Recognition.

## RESPONSIBILITIES

- 5+ years of Machine Learning experience.
- Leading projects to achieve challenging metrics as the Lead Developer.
- Gathering and determining business requirements as the project lead.
- Mentoring team members while personally contributing to multiple projects.
- Ensuring team members stay on task while handling the project.
- Motivating team members by encouraging them and creating an inspiring team environment.
- Providing prompt technical support for the developed products.

## SKILLS

Python

C/C++

Machine Learning

Tensorflow

Keras

AWS

NumPy

Pandas

Scikit-Learn

OpenCV

NLTK

RTI-DDS

RabbitMQ

ROS

Git

Linux

SQL

Data Mining

Data Analytics

## EDUCATION

M.Tech in Information Technology (Machine Learning & Intelligent Systems)

IIIT Allahabad

📅 2017 - 2019

8.42 CGPA

B.Tech in Computer Science

College Of Engineering Roorkee

📅 2012 - 2016

Senior Secondary

CBSE Board

📅 2012

## PROJECTS

- Identity Verification of Indian Aadhaar and PAN Card

Celusion Technologies

📅 2021-2022

Fast and accurate detection of Indian Aadhaar and PAN card from user's uploaded image, and subsequent OCR to automate retrieval of user information which is then cross-checked with user's request form in banking software.

- Identity Verification For Philippines Driver's License

Celusion Technologies

📅 2021-2022

Fast and accurate detection of Philippines Driver license from user's uploaded image, and subsequent OCR to automate retrieval of user information which is then cross-checked with user's request form.

- Face DeDupe

Celusion Technologies

📅 2022

Using K-means clustering and facial landmarks, check any new request form's user image for existence in banking database. If a match is detected but user enters conflicting details, the software warns about possible fraud.

- **Face Liveliness**

**Celusion Technologies**

📅 2021-2022

This project combines facial recognition with facial passive liveliness detection to prevent spoofing of bio-metric samples in banking software.

- **Coastal Surveillance System-Myanmar**

**CRL-B.E.L**

📅 2020-2021

Enabled a seamless integration of the sensors on TCP, UDP and RabbitMQ network, then processed data for sensor data fusion.

- **Integrated Perimeter Security System-India**

**CRL-B.E.L**

📅 2019-2020

Developed a Weapon Detection Module using object detection to improve the surveillance system for real-time Camera Feed.

- **Sensor Planning Deployment**

**CRL-B.E.L**

📅 2019-2020

Developed an Application to determine the minimum number of sensors required for effective surveillance of designated area.

- **Master's Thesis : Localization using Machine Learning in partially Dynamic Environment**

**IIIT-Allahabad**

📅 2018-2019

Proposed and tested methodology within the framework of CNN-LSTM for the real-time localization of a robot in static and dynamic environment. Improved on state-of-the-art accuracy.

- **Self-Driving Car in Simulator**

**IIIT-Allahabad**

📅 2018

Nvidia-based CNN architecture for predicting Steering Angle, Throttle, Brake, Speed which are the basic parameters for driving a car. These predicted parameters are then inferred in real-time for each frame.

- **Human Pose Comparison**

**IIIT-Allahabad**

📅 2018

Using machine learning and computer vision, compare the poses of instructor and student. The goal of this project is to give real-time feedback to the student to improve their form in areas like dance, yoga or other exercises and subsequently, help him/her to understand their areas of improvement.

- **Netflix Movie Recommendation System**

**IIIT-Allahabad**

📅 2018

Recommend favorable movies to users which they have not watched using Machine Learning.

- **Polarity Prediction of given products on Amazon Food Reviews**

**IIIT-Allahabad**

📅 2017

Classified the reviews of various food products as positive reviews or negative reviews using KNN and Naive Bayes.

- **Robot Navigation using Overhead Camera and Reinforcement Learning**

**C.O.E.R**

📅 2015-16

Developed real-time reinforcement learning-based solutions to make the robot move intelligently from the initial position, collect the ball from a location and score the goal.

- **Facial Recognition**

**C.O.E.R**

📅 2015

Using PCA we avoid Curse of Dimensionality for faces which are high dimension data. We then feed the extracted features into an SVM classifier to create a user Classification model.

- **Semi-Automated FAQ-Retrieval System**

**R.W.X Technology**

📅 2015

Developed an FAQ System Using Data mining and NLP that returns the Top-10 best-matching questions for user query quickly.

- **Hand Gesture Recognition**

**R.W.X Technology**

📅 2015

Using SSD model to perform real-time object detection of hand gestures from a finite number of gesture classes in American Sign Language.

## ACHIEVEMENTS

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- Indian national exam GATE percentile 99.05 (rank 1143).
- Machine Learning course by Stanford University on Coursera.
- Applied Machine Learning course by Applied AI.
- Geo-location mapping in Leaflet JS course on Udemy.
- Backend REST API with Python & Django course on Udemy.
- Secured 1st position (Team Captain) in Badminton, RUNN'16 and 2nd position (Team Captain) in Badminton, RUNN'15 of College .
- Secured 2nd position in 6th Inter College UTU (Uttarakhand Technical University) Badminton Tournament 2015-16 in Men's Doubles.
- Played Badminton at National School Level in 2011.